

36. Recognise Watch from the Given Image by General Object Recognition Using OpenCV

Aim

To recognise a watch from the given image using general object recognition techniques in OpenCV.

Procedure

1. Import the required OpenCV and NumPy libraries.
2. Read the given watch image.
3. Convert the image into grayscale.
4. Apply Gaussian blur to reduce noise.
5. Detect circular regions using the Hough Circle Transform.
6. Identify the main circular region as the watch face.
7. Draw a bounding circle around the detected watch.
8. Display the recognised object with the label "WATCH".
9. Save the output image.

Program

```
import cv2

import numpy as np

# Read the input image

img = cv2.imread("watch.jpg")

# Convert image to grayscale

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

```
# Apply Gaussian blur

gray = cv2.GaussianBlur(gray, (5, 5), 0)


# Detect circles

circles = cv2.HoughCircles(

    gray,

    cv2.HOUGH_GRADIENT,

    dp=1.2,

    minDist=50,

    param1=100,

    param2=50,

    minRadius=30,

    maxRadius=100

)


# Check whether a circle is detected

if circles is not None:

    circles = np.uint16(np.around(circles))


# Select the first detected circle

x, y, r = circles[0][0]


# Draw the detected watch face

cv2.circle(img, (x, y), r, (0, 255, 0), 3)
```

```
# Display object name

cv2.putText(

    img,

    "WATCH",

    (x - r, y - r - 10),

    cv2.FONT_HERSHEY_SIMPLEX,

    1,

    (0, 255, 0),

    2

)

print("Object Recognised: WATCH")

else:

    print("Watch not detected")

# Display output

cv2.imshow("Watch Recognition", img)

# Save output

cv2.imwrite("watch_output.jpg", img)

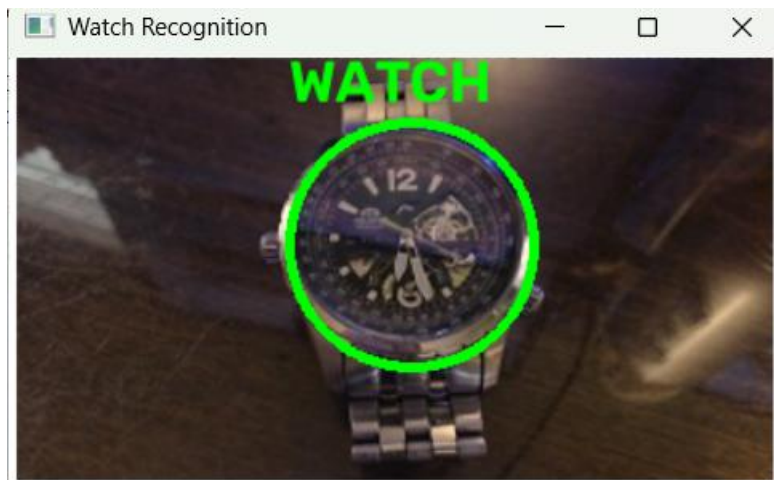
cv2.waitKey(0)

cv2.destroyAllWindows()
```

Input



Output



Result

Thus, the watch was successfully recognised from the given image using OpenCV. The Hough Circle Transform was used to detect the circular watch face and identify it as a WATCH.